As hematopoietic stem cells can give rise to all of the different types of blood cells, they are often known as pluripotent stem cells. (Figure 13.10)

The myeloid progenitor gives rise to erythrocytes, neutrophils, eosinophils, basophils, monocytes, mast cells and platelets. The common lymphoid progenitor gives rise to B lymphocytes, T lymphocytes and natural killer (NK) cells.

13.3.2 Types of Leukocytes

The cells responsible for both innate immunity and acquired immunity are the leukocytes (Greek leukos, white and kytos cell). The average adult has approximately 7400 leukocytes (white blood cells) per cubic millimeter of blood (Table 13.2). The average value shifts substantially during an immune response. In defending the host against pathogenic microorganisms, leukocytes cooperate with each other first to recognize the pathogen as an invader and then to destroy it. The different types of leukocytes are now briefly described.

a. Granulocytes

Granulocytes have irregularly shaped nuclei with two or five lobes. Their cytoplasm has granules that contain reactive substances that kill microorganisms and enhance inflammation. Three types of granulocytes exist: basophils, eosinophils, and neutrophils. Because of the many lobed (3-5) nuclei, neutrophils are also called polymorphonuclear neutrophils or PMNS (Figure 13.11).

HOTS

Heard about stem cell treatment!

Why do we need stem cells bank?

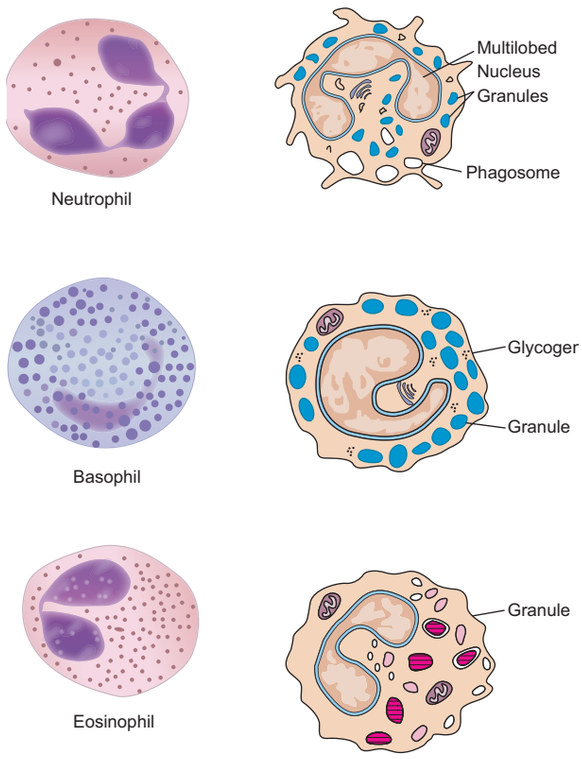


Figure 13.11: Structure of granulocytes

Table 13.2: Normal Adult Blood Count

Cell type	Cells/ mm3	% WBC
Red blood cells	50,00,000	–
Platelets	2,50,000	–
Leukocytes	7,400	100
Neutrophil	4320	60
Lymphocytes	2160	30
Monocytes	430	6
Eosinophils	215	3
Basophils	70	1

b. Mast cells

Mast cells are bone marrow derived cells that differentiate in the blood and connective tissue.

c. Monocytes and Macrophages

Monocytes are mononuclear leukocytes. They are produced in the bone marrow and enter the blood, circulate for about eight

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